

Homework — Week 3, Day 1

Due: **before the next session** (instructor: fill date ____).

Submit in a GitHub repo (same as weeks 1–2, or a new `epic-go-week03` repo). Folder layout:

```
week03/  
  control.go  
  slices.go  
  freq.go  
  slicekit/  
    go.mod  
    slicekit.go  
    slicekit_test.go  
  exercises.md      # short answers for 1–8
```

No zip files, no WhatsApp screenshots-only.

Exercises 1–8 (write the answer in `exercises.md`)

Run the code. Then write **one or two sentences** plus the output. Your words.

1. **if short statement.** Write an `if` that declares `n := len(s)` and prints `"empty"` when `n == 0`. Is `n` visible after the `if`?
2. **Tagless switch.** Grade bands for scores 80 / 65 / 40 using a tagless `switch`. Paste the code.
3. **for shapes.** Show a C-style loop and a while-style loop that both print `3 2 1`.
4. **append assign.** Start with `var xs []int`, append `1`, then `2`. What is wrong with calling `append` and ignoring the return?
5. **Backing array.** Reproduce the gotcha: `a := []int{10,20,30,40}`; `b := a[:2]`; `b = append(b, 99)`. What is `a` after? Why?
6. **Independent copy.** Make a copy of a slice that does **not** share a backing array. Prove it by mutating the copy.
7. **Comma-ok.** On a map, distinguish “key missing” from “key present with value `0`”. Paste the check.
8. **Map range order.** Range the same map twice. Did the print order match? What does that mean for tests?

Exercise 9 — harden the lab package

In `slicekit`, keep green tests for:

```
func Contains(list []string, want string) bool
func Unique(list []string) []string
func Freq(words []string) map[string]int
```

Add these cases (table-driven is fine; copy the week-2 test style if you prefer):

Function	Input	Want
<code>Contains</code>	<code>nil, "SE"</code>	<code>false</code>
<code>Unique</code>	<code>["SE", "CS", "SE", "IT", "CS"]</code>	<code>["SE", "CS", "IT"]</code> (order preserved)
<code>Unique</code>	<code>nil</code> or empty	empty slice
<code>Freq</code>	<code>["a", "b", "a"]</code>	<code>a:2, b:1</code>
<code>Freq</code>	empty	empty map (len 0)

```
cd slicekit
go test
```

Must be green. Assert map contents with comma-ok — do **not** compare `fmt.Sprintf` of a map.

Exercise 10 (optional reading)

Skim [Effective Go](#) sections on slices and maps. Write five lines in `exercises.md`: one thing you already knew, one thing that surprised you.

How it is checked

Requirement	Pass
Repo URL sent	Yes
Demo files run	Shown in class or committed
<code>go test</code> in <code>slicekit/</code> is green	Including harden cases
<code>exercises.md</code> is your words	Oral question next class

Requirement	Pass
<code>gofmt</code> clean, no binaries	<code>.gitignore</code> working